A Socratic council operationalizes vague research questions into falsifiable, instrument-testable hypotheses

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Abstract

Autonomous "AI scientist" systems can generate ideas and run experiments, but they treat hypothesis *formulation* as a single generative step and optimize for novelty rather than testability—so their hypotheses are often novel-sounding yet not operationalized against any specific instrument. We introduce the **Socratic Council**, a multi-agent architecture that makes the philosophy of science executable and performs the missing *operationalization* stage: a maieutic *proposer* (abduction), an elenctic *skeptic* (Socratic ignorance), and a rubric-bound *judge* debate a vague question into a falsifiable, discriminating, instrument-testable hypothesis, terminating on a falsifiability criterion rather than on agreement. Evaluated on ten literature-grounded questions in single-cell *E. coli* biology against a whole-cell simulator, the dialectic does not change how *testable* hypotheses *look*—a coarse operationalization-quality rubric saturates for every configuration—but it makes them measurably *sounder*: a fresh, configuration-blind, cross-family auditor finds the full Council leaves ∼20% fewer substantive methodological defects than a single strong prompt or a critic-free ablation (paired one-sided Wilcoxon $p = 0.001$), with the reduction largest on the hardest-to-operationalize questions and the elenctic *skeptic* identified as the active ingredient. A first-class `Hypothesis` object carries an executable falsifier that we run against fresh simulations to return confirm/refute verdicts. An information quarantine—an import-level control that denies the Council the answer key—ensures the metric measures *derivation*, not recall. We argue that a disciplined operationalization stage, held to a philosophy-of-science rubric, is a missing and separable component of AI-driven scientific reasoning.

1 Introduction

A whole-cell simulator or a robotic laboratory can, in principle, test almost any single-cell hypothesis—but only once the hypothesis is stated *testably*. Users arrive with questions such as "do genetically identical cells behave differently?", which name no observable, no baseline, no prediction and no refuting result. Turning such a question into a falsifiable claim is the step that philosophy of science calls *operationalization* (Bridgman, 1927) within Reichenbach's *context of discovery* (Reichenbach, 1938), and it is precisely the step current AI-scientist systems skip.

The dominant systems—Sakana's AI Scientist (Lu et al., 2024; Yamada et al., 2025), Google's AI co-scientist (Gottweis et al., 2025), autonomous chemistry agents (Boiko et al., 2023), the Robot Scientists (King et al., 2004, 2009), and virtual/agent laboratories (Swanson et al., 2024)—automate the loop from ideation to experiment, but treat hypothesis formulation as a lightweight prompt and optimize downstream signals: perceived novelty, or an Elo tournament of expert preference. A large blinded study finds LLM-generated ideas are rated *more novel but less feasible* than expert ideas (Si et al., 2024); ideation systems explicitly optimize novelty (Wang et al., 2024). None imposes an explicit constraint that a hypothesis be *operationalized against a named instrument's measurable capabilities*, be falsifiable in the sense of Popper (1959) and Platt (1964), and be produced under conditions that prevent the answer from leaking into its own formulation.

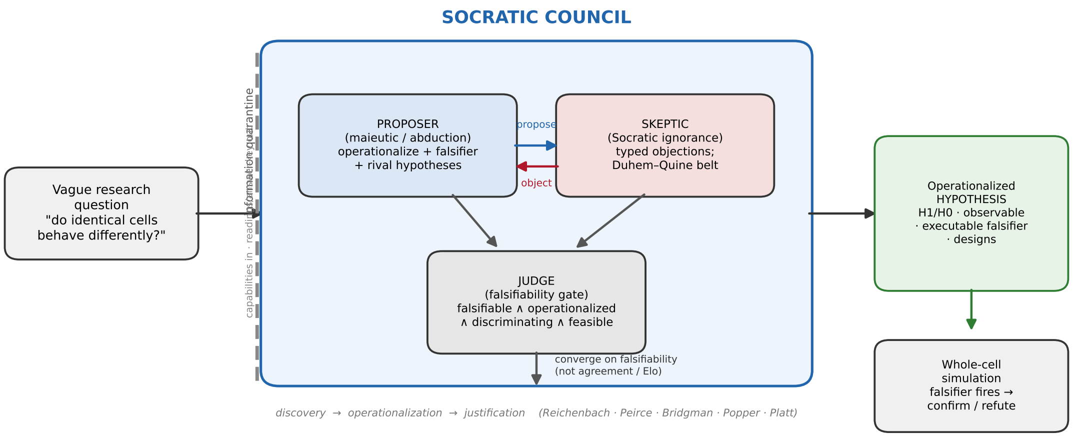

Figure 1. Fig. 1 | The Socratic Council. A vague question is transformed by a proposer (maieutic abduction), an elenctic skeptic (Socratic ignorance), and a rubric-bound judge that converges on falsifiability rather than agreement, yielding a first-class operationalized `Hypothesis` whose executable falsifier is run against a whole-cell simulation. The Council operates behind an *information quarantine*: it sees the instrument's measurable capabilities but never its readings or the literature answer key. The three roles compute Reichenbach's discovery→justification arc via the middle bridge of operationalization.

Multi-agent debate improves factuality and reasoning (Du et al., 2023; Irving et al., 2018), and rubric-bound self-critique is established (Bai et al., 2022; Madaan et al., 2023; Shinn et al., 2023); but these optimize *answer correctness*, not the epistemic quality of a *hypothesis*, and terminate on agreement rather than on falsifiability. We ask whether a dialectic that instantiates the norms of the philosophy of science—abduction, falsifiability, operationalism, strong inference, multiple working hypotheses (Chamberlin, 1890), and the Duhem–Quine belt of auxiliary assumptions (Quine, 1951)—can perform this missing operationalization stage, and whether doing so measurably improves the hypotheses over a single strong prompt.

Contributions. (i) The **Socratic Council**, a proposer/skeptic/judge architecture that computes the discovery→operationalization→justification arc and terminates on a falsifiability gate. (ii) A pre-registered evaluation on ten literature-grounded questions showing that the dialectic's value is *not* visible on a coarse quality rubric (which saturates) but is significant and cross-family-replicated on a sensitive residual-defect metric, with the elenctic skeptic as the active ingredient. (iii) A first-class executable `Hypothesis` object and a verdict engine that runs the Council's own falsifiers against fresh whole-cell simulations. (iv) An **information quarantine**: an auditable, import-level control that ensures the system *derives* rather than *recalls* operationalizations.

2 Results

2.1 The Socratic Council

The Council (Fig. 1) transforms a question through three roles, each an epistemic stance. The **maieutic proposer** performs abduction: it emits the sharpest candidate hypothesis, operationalizing every construct onto a real instrument observable (Bridgman), with a predicted effect, a falsifier (a risky prohibition that could fail; Popper), and at least two rival hypotheses each with a distinguishing experiment (Chamberlin/Platt). The **elenctic skeptic** adopts Socratic ignorance: it raises typed objections (undefined term, hidden Duhem–Quine auxiliary, unfalsifiable claim, rival-not-excluded, or a claim that outruns the instrument). The **rubric-bound judge** is a gate, not a scorer: it certifies a hypothesis only when it is falsifiable, specified, operationalized onto real observables with a named statistical test, and discriminating, and terminates on this falsifiability criterion together with a convergence signal—never on mere agreement, novelty, or Elo. The output is a first-class `Hypothesis` object: H_1/H_0, operational definitions bound to observables, an executable falsifier, rival hypotheses, the auxiliary-assumption belt, and envelope-checked candidate experimental designs.

A worked trace on the flagship question "do genetically identical *E. coli* cells behave differently?" illustrates the mechanism. The proposer produces a hypothesis (isogenic cells show phenotypic heterogeneity, coefficient of variation ≥ 0.15 across replicate seeds, driven by stochastic gene expression) with a CV-based falsifier and three rivals. In a second round the skeptic identifies two *substantive* defects a single-shot prompt would have shipped: a control design that "outruns the instrument" (it presumes a perturbation can suppress variance when it can only shift the mean), and a rival-discrimination prediction that is internally *contradictory.* The proposer repairs both—replacing the infeasible control, clarifying the mechanism, and adding a second knockout as a discriminating arm—after which the judge certifies convergence. The dialectic thus improves the *experimental design*, not merely the prose.

2.2 The dialectic makes hypotheses sounder, not more testable-looking

We evaluated four configurations—**full** (proposer+skeptic+judge), **no_skeptic** (proposer+judge), **proposer_only** (single-shot generation, the baseline), and **generic_judge** (proposer+skeptic with a plain quality judge, no falsifiability rubric)—on ten literature-grounded questions (three replicates each; Claude Sonnet 4.5 roles at temperature 0.7). Two metrics were pre-registered: a six-criterion operationalization-quality rubric graded blind to the literature answer, and—after observing the rubric saturate on the first cases, and before the full audit was read—a more sensitive *residual-defect* count from a fresh adversarial auditor blind to configuration. Both metrics were scored by an independent Claude judge *and* a cross-family GPT-4o judge.

The binary quality rubric is a clean **null** (Fig. 2C): every configuration scores $\approx 6/6$ (full and proposer_only both 6.0; Claude–GPT mean absolute score difference 0.017). A single strong proposer already writes *formally* adequate hypotheses; the dialectic does not change how testable they *look*.

On the sensitive metric, the full Council leaves **fewer substantive methodological defects** than the baselines (Fig. 2A). On the cross-family GPT auditor, the full Council averages 3.80 defects versus 4.77 for single-shot and 4.67 for no_skeptic—a $\sim$20% reduction, significant under a one-sided paired Wilcoxon signed-rank test across the ten cases ($p = 0.001$ for both comparisons). The independent Claude auditor agrees in direction (3.37 vs 4.50 and 4.83, a $\sim$25% reduction) though with wider case-to-case variance and without significance. The reduction is **largest on the hardest-to-operationalize questions** (Fig. 2B): 2.0 fewer defects on the flagship heterogeneity question and 1.3 on non-genetic individuality, versus ≈ 0.3 on well-trodden ppGpp physiology where a single prompt already suffices. Every one of the ten cases favours the full Council, which is why the paired test is significant despite modest per-case effects.

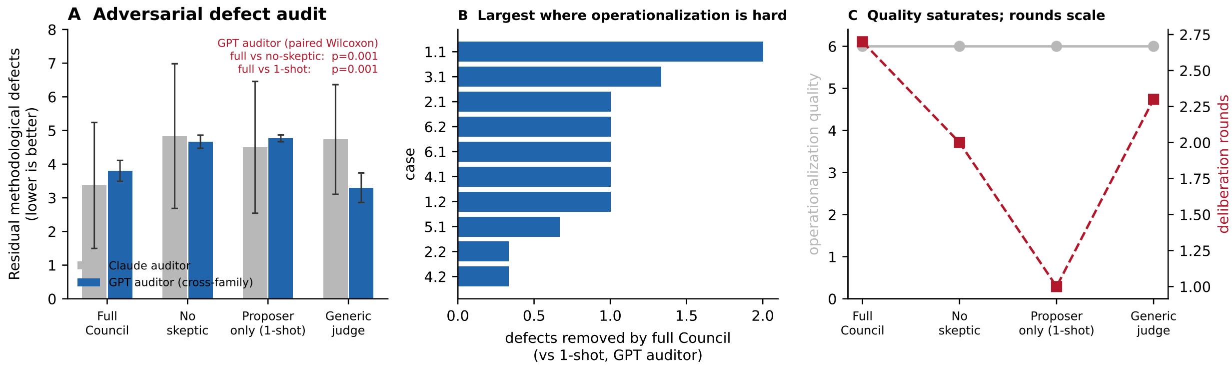

Figure 2. Fig. 2 | The Socratic dialectic makes hypotheses sounder, not more testable-looking. **A**, Residual methodological defects per configuration, counted by a configuration-blind adversarial auditor (grey: Claude Opus 4.8; blue: cross-family GPT-4o); error bars are 95% case-clustered confidence intervals. The full Council leaves fewer defects than the critic-free ablation and the single-shot baseline (cross-family auditor, one-sided paired Wilcoxon, $p = 0.001$). **B**, Per-case reduction of the full Council versus single-shot (cross-family auditor); the effect is largest on the questions hardest to operationalize (1.1 isogenic heterogeneity; 3.1 non-genetic individuality) and smallest on well-trodden physiology (4.2 ppGpp). Every case favours the full Council. **C**, The two supporting signals: the binary operationalization-quality rubric saturates (grey, a null—the dialectic does not change how testable hypotheses look) while deliberation rounds scale with the dialectic (red). $n = 3$ replicates $\times$ 10 cases per configuration.

2.3 The elenctic skeptic is the active ingredient

Component isolation localizes the effect to the critic rather than the rubric. `generic_judge`, which retains the skeptic but discards the falsifiability rubric, matches the full Council on the cross-family defect metric (3.30 vs 3.80), whereas `no_skeptic` is clearly worse (4.67). Deliberation depth scales correspondingly (Fig. 2C): mean rounds are 2.7 (full), 2.3 (generic_judge), 2.0 (no_skeptic) and 1.0 (single-shot). The dialectic's cost is more deliberation; its benefit is the removal of methodological defects a single pass leaves behind.

2.4 The hypotheses are instrument-testable

The `Hypothesis` object's falsifier is executable. A verdict engine compiles it into a structured test (the statistic, channel, target and reference designs, and threshold) and runs the actual statistic on per-seed whole-cell simulation output, returning a confirm/refute verdict with a severity assessment against a pre-registered smallest effect size of interest—not post-hoc power (Mayo, 2018). On the flagship question, executed against $n = 16$ real isogenic simulations, the across-seed coefficient of variation is 0.066 for growth rate (H_1 refuted, below the 0.10 threshold), 0.013 for ribosome concentration (refuted), and 0.112 for ppGpp concentration (supported, exceeding the effect-size floor). The loop closes: a vague question becomes an operationalized hypothesis, whose falsifier fires on fresh simulation output and returns a verdict.

We stress the epistemic scope (Sec. 4): a verdict against a whole-cell model is a claim about the *model*, which was itself fit to much of this literature (Macklin et al., 2020; Oreskes et al., 1994); for the in-canon questions the loop is best read as *blinded rediscovery / operationalization validation*, not empirical confirmation (Winsberg, 2010).

2.5 The quarantine measures derivation, not recall

Because the proposer is a large language model that has read this literature, the evaluation must distinguish a *derived* operationalization from a *recalled* answer. We enforce this with an **information quarantine**: the module exposing the instrument to the Council is an import-level boundary that provides the instrument's measurable capabilities (channel names, the runnable-perturbation vocabulary) but never its readings or the literature answer key, and is statically prevented from importing any result-bearing code. In scoping this control we found a real leak—a corpus statistic embedded in a tool's textual description—which the boundary now excludes; the quarantine is thus an auditable engineering control against evaluation contamination, of independent interest for closed-loop AI-science systems.

A leaked-versus-quarantined ablation (the proposer additionally handed each case's canonical answer; $n = 3$ replicates $\times$ 10 cases, both auditors) shows what leakage actually changes. It does *not* consistently improve methodological soundness: the residual-defect count moves in opposite directions under the two auditors (Claude 2.8 leaked vs 4.7 quarantined; GPT 4.1 vs 3.3). What it changes is *recitation of measured values the model would otherwise not assert.* The clearest case is essentiality (5.1), whose canonical answer is a specific count ($\sim$300–620 essential genes): the leaked Council recites that count in 2/3 runs, whereas the quarantined Council recites it in 0/3, proposing instead a *falsifiable fraction* to test. Where the value is derivable or textbook-familiar (the well-known five-fold ppGpp rise, 4.2), both recite it and the quarantine neither can nor should matter. A benchmark scoring "matches the literature" can thus be inflated by recitation, which the quarantine—by construction—prevents; our operationalization-quality metric is therefore an honest measure of derivation. Whether leakage additionally drives a *wrong* downstream conclusion on fresh simulations, rather than mere recitation, requires executing the leaked hypotheses through the verdict engine and is left to future work.

3 Discussion

Our central finding is deflationary in a useful way: a philosophy-of-science dialectic does not make an LLM's hypotheses *appear* more testable—on a coarse rubric, a single strong prompt already writes falsifiable, discriminating hypotheses—but it makes them *sounder*, removing substantive methodological defects that a single pass leaves behind, most where operationalization is hardest, with the elenctic critique as the active ingredient. This is a claim about *operationalization*, a component that current AI-scientist systems subsume into a novelty-optimizing ideation step. Isolated and held to a rubric, it is separately improvable and separately measurable.

Two limitations bound the claims. First, evaluation is by LLM judges; we mitigate with a configuration-blind adversarial auditor, a cross-family replication (the significant result is on the non-Claude auditor), and an answer-key quarantine, but human-expert validation and additional families remain necessary. Second, the instrument is a single simulator fit to the canon: in-canon closed-loop verdicts are rediscovery, not confirmation, and only a pre-registered out-of-canon or wet-lab test would license a discovery claim. Finally, we resist the temptation to over-read the Socratic framing: the skeptic checks the candidate against an external rubric rather than deriving a contradiction from the interlocutor's own commitments, so the elenchus is an organizing analogy (Vlastos, 1983) whose value we establish empirically, not by etymology.

The transferable message for autonomous science is that operationalization deserves to be a first-class, instrumented stage—with a falsifiability-based termination criterion and an information quarantine—rather than a side effect of ideation. A hypothesis that is novel but not operationalized cannot be tested; the Council is a step toward closing that gap.

4 Methods

The Council. Roles are large-language-model calls with forced-tool structured outputs (Claude Sonnet 4.5, temperature 0.7). The judge applies a conjunctive gate over {falsifiable, specified, operationalized, discriminating, feasible}; feasibility is computed deterministically by checking each candidate design against the simulator's validated-perturbation envelope and a biosecurity screen. Termination requires a clean judge certification plus a convergence signal (no new substantive objection, or two stably-clean rounds); the operative early-exit is stability, not a fixed quota. On an irreducible construct ambiguity the Council queries the user.

Instrument and quarantine. The substrate is the public Covert-lab whole-cell *E. coli* model (Macklin et al., 2020); the Council sees only a capability adapter (summary-channel names and units, species kinds, the validated-perturbation vocabulary, the reference design, and the falsification mechanism) that imports no result-bearing module and reads nothing under the data directory. A unit-test asserts the boundary.

Evaluation. Ten questions span intrinsic/extrinsic noise, persistence, bet-hedging, non-genetic individuality, growth laws, the stringent response, essentiality and diauxie, each with a canonical answer, excluded rivals, and a wcEcoli observable mapping (Extended Data). The operationalization-quality rubric has six answer-independent criteria; the residual-defect audit counts substantive methodological defects (falsifier-cannot-fail, rival-not-discriminating, internal contradiction, infeasible design, unstated auxiliary, vague construct, under-scoped) found by an auditor blind to configuration. Each hypothesis is graded by an independent Claude (Opus 4.8) judge and a cross-family GPT-4o judge. The primary comparison is the full Council versus single-shot on the case-clustered mean defect count (mean over three replicates within a case, then across ten cases), by one-sided paired Wilcoxon signed-rank; convergence and feasibility rates use Wilson intervals. Sampling temperature is pinned for the Council roles (the named variance source); reasoning-model graders run at their defaults. Pre-registration (endpoints, rubric, per-case severity thresholds) is committed before the confirmatory results were read.

Verdict engine. A falsifier is compiled into a structured spec and executed on per-seed simulation output: for a dispersion test, the across-seed coefficient of variation against an absolute threshold (the simulator is seed-deterministic, so there is no same-seed technical-noise null); for a difference test, a Welch t on design means. Verdicts are scored target-versus-reference from the runs, not against the pooled corpus. Severity is reported as whether the executed effect exceeds a pre-registered smallest effect size of interest.

Data and code availability. All code, the ten-case benchmark, the pre-registration, and every result JSON are in the project repository; figures regenerate from the committed result files.

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